

Mousetrap Car Design Project

Final Project Report for EE4000 Mechanical Engineering, G Block

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Introduction:

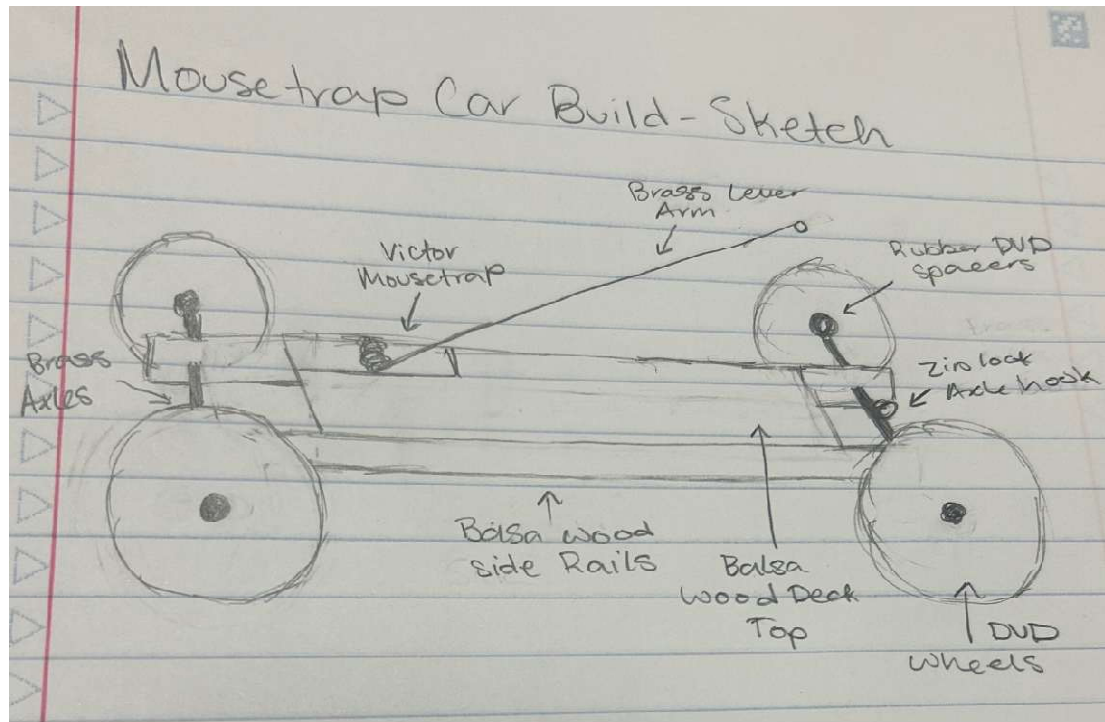
The basic mouse-trap car consists of a wooden frame, four discs as wheels, two short rods connected to the wheels, and a large rod attached to a string that drives the car by winding up the wheels. Following data collection on mechanical advantage, mass, and wheel circumference, one major problem was identified. The car does not cover a great distance. Based on this conclusion, the goal is to enhance the overall distance the mousetrap car can cover by maximizing energy efficiency, minimizing friction, and improving specific parts of the car.

The project will show that a 10% increase in distance is possible by changing the back wheels-, material of the cars body, axel system and length of the lever arm of the mouse-trap car. Each modification to the car was applied to the basic mouse-trap car model, and testing was done to determine if that modification improved the distance the car was able to cover. These modifications include wheel configuration changes, prioritizing weight reduction, adjustments to lever length, and new body dimensions. Then, based on the performance of the car, certain modifications were applied to the final car.

Due to the fragility of the basic mouse-trap car, the data collection on certain modifications was difficult. When testing was possible, mechanical advantage of systems such as the levers, wheels and axles, was noted. In addition, overall distance was measured. However, most of the testing was focused on comparisons between the basic mouse-trap car and the final modified car.

Mousetrap Car Build

The mousetrap car build process mainly involved following step by step instructions provided by Doc Fizzix. Materials included two pre-drilled balsa wood side rails (18" * 1 1/2"), one balsa wood deck top (12" * 3"), two brass axles (4" * 3/16" tube), one brass lever arm (12" * 1/8" tube), four DVD wheels, four rubber DVD/CD spacers, four metal thrust washers, one victor mousetrap, one zip lock axle hook, and 36" Kevlar string. Due to the ease of instructions provided, there were no issues in building the basic mousetrap car.



Initial Mousetrap Car Analysis & Test

Following the construction of the basic mouse-car trap model, testing and analysis was done to collect baseline data, which could be used to compare with the final mouse-trap car.

First, work done by the mousetrap arm was found. This calculation was done using the formula: $Work (W) = Force (F \text{ in newtons}) \times distance (d \text{ in meters})$. By measuring the length of the mousetrap arm, 32cm, and using the formula for circumference, which is $2\pi r$, it was determined that the mousetrap arm tip would travel 201 cm if it made an entire rotation. However, the maximum distance traveled is only half of a circle due to the way the mouse-trap car is constructed, so the maximum distance is only 100.5cm, or 1.005m. The average force when pulled back. If the distance is 1.005m, and the average force is 1.0N, which was measured with a spring scale, then the work done by the arm is 1.005 Joules when the arm is pulled back 100%.

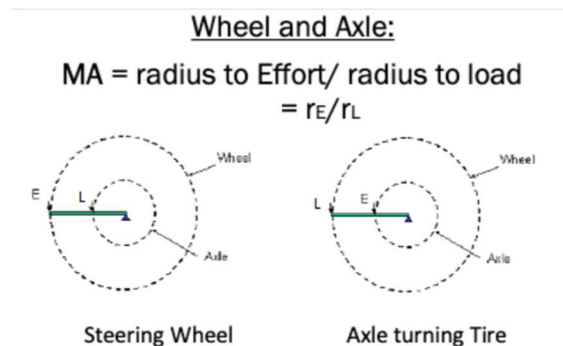
Second, the potential energy was measured. The spring's tension was measured at 10-degree intervals using a protractor, and the average potential energy was calculated for each angle. The total potential energy of the mousetrap was found by adding all values.

Angle (degrees) <i>Position on protractor</i>	Force (Newtons) <i>Measured with Spring Scale</i>	Average Force (Newtons) <i>Sum of Previous Force and This Force</i>	Angular Distance (d) (meters) <i>Arc Radius * Angle * pi/ 180</i>	Average Potential Energy (Joules ⁺) <i>Average Force Value * Angular Distance Traveled⁺⁺</i>
0	F1 = 0	-----	-----	-----
10	F2 = 0.6 N	F1-2 = (F1 + F2)/2 0.3 N	d = 0.055m	PE = F1-2 x d = 0.0165
20	F3 = 0.67 N	F2-3 = (F2 + F3)/2 0.635 N	d = 0.11m	PE = F2-3 x d = 0.0698
30	F4 = 0.68 N	F3-4 = 0.675 N	d = 0.16m	PE = F3-4 x d = 0.108
40	F5 = 0.7 N	F4-5 = 0.69 N	d = 0.22m	PE = 0.1518
50	F6 = 0.72 N	F5-6 = 0.71 N	d = 0.27m	PE = 0.1917
60	F7 = 0.76 N	F6-7 = 0.74 N	d = 0.33m	PE = 0.2442
70	F8 = 0.8 N	F7-8 = 0.78 N	d = 0.39m	PE = 0.3042
80	F9 = 0.81 N	F8-9 = 0.805N	d = 0.44m	PE = 0.3542
90	F10 = 0.84 N	F9-10 = 0.825 N	d = 0.50m	PE = 0.4125
100	F11 = 0.62 N	F10-11 = 0.73 N	d = 0.55m	PE = 0.4015
110	F12 = 0.67 N	F11-12 = 0.645 N	d = 0.61m	PE = 0.3934
120	F13 = 0.73 N	F12-13 = 0.7 N	d = 0.67m	PE = 0.469
130	F14 = 0.76 N	F13-14 = 0.745 N	d = 0.72m	PE = 0.5364
140	F15 = 0.78 N	F14-15 = 0.77 N	d = 0.78m	PE = 0.6006
150	F16 = 0.83 N	F15-16 = 0.805 N	d = 0.83m	PE = 0.6681
160	F17 = 0.9 N	F16-17 = 0.865 N	d = 0.89m	PE = 0.7698
170	F18 = 0.95 N	F17-18 = 0.925 N	d = 0.94m	PE = 0.8695
180	F19 = 1.1 N	F18-19 = 1.0 N	d = 1.0m	PE = 1.0125

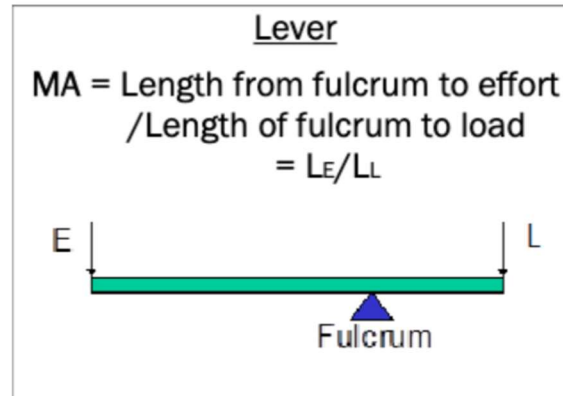
Total PE⁺⁺⁺: 7.5737 J

Third, the mechanical advantage of the wheels and axles, and the levers, was calculated.

Calculating the Mechanical Advantage of a Wheel and Axle System:



Calculating the Mechanical Advantage of Levers:



The formula for the mechanical advantage of a wheel and axle system is the radius of the wheel divided by the radius of the axle. For the basic-mouse trap car, the radius of the wheel was 4cm, or 0.04m, and the radius of the axle was 0.25cm or 0.0025m. Using the formula for mechanical advantage, the equation becomes 0.04m/0.0025m, which produces a mechanical advantage of 16. The mechanical advantage of the lever was found by measuring the distance of the small, curved arm on the spring and the length of the lever from the base of the mousetrap spring to the loop where the string will be tied. The length of the lever from the base of the car to the loop is 32cm, and the length of the small, curved arm on the spring is 4.25cm. Using the formula, 32 cm is divided by 4.25 cm, which means the lever of the basic mouse trap car produces a mechanical advantage of 7.5.

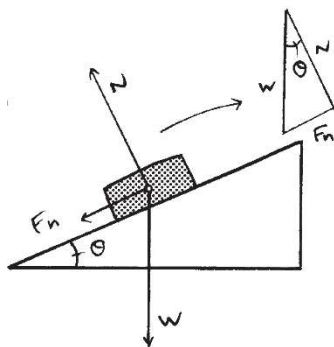
Forth, data was collected about the wheel circumference provides information about the number of revolutions the wheels make.

Radius (cm) r	Diameter (cm) d	Circumference $C = \pi * d$
4cm	8cm	25.13cm

Distance d	Circumference (cm) C	Number of Revolutions $Rev = d/C$
100 centimeters	25.13cm	$Rev = 100cm / 25.12cm$ $Rev = 3.98cm$
100 inches	9.89in	$Rev = 100in / 9.89in$ $Rev = 10.11in$
200 centimeters	25.13cm	$Rev = 200cm / 25.13cm$ $Rev = 7.96cm$
200 inches	9.89in	$Rev = 200in / 9.89in$ $Rev = 20.22in$
10 meter (1000 centimeters)	0.2513m 25.13cm	$Rev = 10m / 0.2513m$ $Rev = 39.8m$
100 feet (1,200 inches)	0.82ft 9.89in	$Rev = 100ft / 0.82ft$ $Rev = 122ft$

Finally, the rolling friction was calculated. This was done to determine the amount of rolling friction acting against your mousetrap car and the coefficient of friction.

Calculating the Rolling Friction:



Friction is a force that acts against the motion of all moving objects. The less friction acting against a moving mousetrap car, the less energy that is consumed to friction and further that the vehicle will travel.

A moving mousetrap car is affected by two types of friction: air friction and bearing friction. To find the theoretical rolling friction requires placing your mousetrap car on a smooth and flat board or ramp. To calculate the theoretical rolling friction, a ramp is elevated from one end slowly until the mousetrap car begins rolling at a constant velocity. Using this data, the coefficient of friction can be found, as well as the force of friction. T

Rolling Friction Data Table #1

Trial #	Board Length (m)	Raised Height (m)	Angle	Coefficient of Rolling Friction	Friction(N)	Starting Energy (J)	Predicted Travel Distance
1	L= 84.5cm 0.845m	h1= 1 cm 0.01 m	$\theta_1 = 0.011$	$\mu_1 = 0.01$	$f_1 = 0.098$	PE= 0.0981	d1=1
2	L=84.5 cm 0.845m	h2= 3 cm 0.03m	$\theta_2 = 0.035$	$\mu_2 = 0.03$	$f_2 = 0.293$	PE= 0.2943	d2=1
3	L=84.5 cm 0.845m	h3= 3 cm 0.03m	$\theta_3 = 0.035$	$\mu_3 = 0.03$	$f_3 = 0.293$	PE= 0.2943	d3=1
4	L=84.5 cm 0.845m	h4= 4 cm 0.04cm	$\theta_4 = 0.047$	$\mu_4 = 0.04$	$f_4 = 0.3896$	PE= 0.3924	d4=1.01
AVE	0.845	-	$\theta = 0.128$	$\mu =$	$f =$		d=1

To calculate the force of friction by assuming that the force down the ramp is equal to the force of friction acting against the motion of the vehicle. Using the equation weight equals mass times gravity (9.8 m/s^2) as well as the potential energy calculated earlier, the predicted travel distance can be found by using the following formula: (Predicted Travel Distance = Total Potential Energy (joules or N-m)/(Rolling Friction))

$$W = m \times g$$

$$W = 0.1231\text{kg} \times 9.8\text{m/s}^2 \approx 1.206\text{N}$$

$$F = W \times \sin(\theta)$$

$$F = 1.206\text{N} \times \sin(30) = 1.206 \text{ N} \times 0.5 \approx \mathbf{0.603\text{N}}$$

$$\text{PE} = 0.1231\text{kg} \times 9.8\text{m/s}^2 \times 1.0\text{m} \approx 1.206 \text{ J}$$

$$\text{Predicted Travel Distance} = 0.603\text{N}/1.206\text{J} \approx \mathbf{2.00\text{m}}$$

Design Project

Engineering Design Process.

The engineering design process was essential in guiding our development of an optimized mousetrap car capable of maximizing distance and energy efficiency. By following each step of the process, we identified specific challenges, brainstormed innovative solutions, and systematically refined our design through testing and iteration. The structured approach helped break down complex problems into manageable steps, ensuring that every modification aligned with our performance goals.

Problem Definition

The goal of this process is to enhance the overall distance the mousetrap car can cover by maximizing energy efficiency, minimizing friction, and improving specific parts of the car.

This requires changing certain aspects of the design, such as the lever arm length, wheel size, and weight distribution, while ensuring that the car remains stable and efficient.

Now that we have defined the problem and goal, we will research and brainstorm solutions by reviewing factors that influence the car's performance like energy transfer and aerodynamic efficiency. Next, to develop potential solutions, we are thinking of increasing the lever arm length to store more energy, using lighter and less materials for the car body, reducing wheel friction by using smoother bearings or larger wheels, reducing the diameter of the driving rod and adjusting the weight distribution for stability. When we prototype and test the changes we make, we will make modifications to the car and test these changes to determine the impact on the distance it travels. After testing, we will analyze the data to determine which modifications resulted in the most significant performance improvements. The final step will be to iterate and refine the design, meaning we will reflect on the testing and make the necessary changes on the car to have the most efficient car.

Research Problem

When researching the problem, some improvements others had made were compiled. These improvements included increasing the lever arm length by extending the lever arm, reducing friction by using lightweight, low-friction materials for the axles to help minimize energy lost to friction, finding larger rear wheels increase the car's travel distance by covering more ground with each rotation, and changing the car's shape to a more aerodynamic design to reduce air resistance.

Using this information, the group of engineers decided that to improve the car's performance possible solutions included increasing the lever arm, using larger wheels, minimizing friction by making the wheel's smoother, improving aerodynamics by planning to use a sleeker frame design longer than the current frame.

Requirements

1. *Increase the car's travel distance by at Least 50%*
2. *Optimize energy transfer with at least 25% efficiency*
3. *Decrease overall weight by 20%*
4. *Increase the duration of energy release by 30%*
5. *Ensure stability throughout the Run (no veering off course, tipping over, or wobbling)*
6. *Minimize energy loss due to resistance and friction, ensure less no more than a 5% loss of energy*

Brainstorming

Final set of requirements agreed upon as a group:

1. In order to increase the distance of the car we will increase the size of the lever arm and make it longer

Calculations need to be done to find optimal length

2. In order to cover more distance we will use larger CDs/3D printed wheels for the wheels
Holes in the wheels to reduce weight

Two large wheels in the back

Two small wheels in the front/one small wheel in the front

3. In order to improve aerodynamics we will use a sleeker frame design that is much longer than the current frame.
4. In order to make the car lightweight, we will shave away as much material as possible
5. Decrease the diameter of the axle to increase rpm for the
 - a. Make axle as short as possible to avoid problems/waste energy
6. Attempt to make structure of car resemble a teardrop for the best aerodynamics

Decision Matrix and Selection

		Total Weight	10.8
Weight	Criteria	Rank	% of decision
1.5	Increase size of lever arm	4	28%
3	Change Wheel Configuration	2	19%
2.5	Sleeker Frame/Teardrop Structure (Aerodynamic)	3	18%
1	Decrease Diameter of Axle	5	15%
2.8	Decrease Mass by Removing Material	1	12%

Weights	CRITERIA					Results		
	3	2	1.9	1.6	1.3	1	RAW SCORE	RANK
	Removes a significant amount of plastic	No Harm to Ocean Life	Sustainable	Achievable with Current Technology	Cost	Implementation Speed		
OPTIONS							WEIGHTED SCORE	
Autonomous Su	7	7	9	10	6	8	47	83.9
Boat Cleaner	6	8	8	10	7	8	47	82.3
Plastic Eating Ba	9	1	6	5	8	6	35	64.8
Man-Made Gyres (Con	5	3	2	4	2	3	19	36.8

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Prototype, Test, Analyze, and Iterate

Data/Results:

Following the construction of the new, modified mouse-car trap model, testing and analysis was done, which could be used to compare with the basic mouse-trap car.

Work done by the mousetrap arm was found. This calculation was done using the formula: Work (W) = Force (F in newtons), * distance (d in meters). By measuring the length of the mousetrap arm, 41cm, and using the formula for circumference, which is $2\pi r$, it was determined that the mousetrap arm tip would travel 257 cm if it made an entire rotation. However, the maximum

distance traveled is only half of a circle due to the way the mouse-trap car is constructed, so the maximum distance is only 128 cm, or 1.285 m. The average force when pulled back. If the distance is 1.285 m, and the average force is 1.0N, which was measured with a spring scale, then the work done by the arm is 1.29 Joules when the arm is pulled back 100%.

The potential energy for the modified mouse-car trap was not calculated, since the spring had not been changed so the data is the same and calculations are not needed.

The mechanical advantage of the wheels and axles, and the levers, was calculated. The formula for the mechanical advantage of a wheel and axle system is the radius of the wheel divided by the radius of the axle. There were two different types of wheels on the modified mouse-trap car, one with a radius of 4cm, or 0.04m, and one with a radius of 8cm. The radius of the axle was 0.25cm or 0.0025m. Using the formula for mechanical advantage, the equation becomes $0.04\text{m}/0.0025\text{m}$ for the front wheels, and $0.08\text{m}/0.0025\text{m}$ for the back wheels, which produces a mechanical advantage of 16 and 32 respectively.

The mechanical advantage of the lever was found by measuring the distance of the small, curved arm on the spring and the length of the lever from the base of the mousetrap spring to the loop where the string will be tied. The length of the lever from the base of the car to the loop is 41 cm, and the length of the small, curved arm on the spring is 4.25cm. Using the formula, 41 cm is divided by 4.25 cm, which means the lever of the basic mouse trap car produces a mechanical advantage of 9.64.

Forth, data was collected about the wheel circumference provides information about the number of revolutions the wheels make. There are sets of wheels, with different measurements, so there will be two data sheets. The first data sheet is the smaller front wheel, and the second data sheet is the larger back wheel.

Radius (cm) r	Diameter (cm) d	Circumference $C = \pi * d$
4cm	8cm	25.13cm

Distance d	Circumference (cm) C	Number of Revolutions $\text{Rev} = d/C$
100 centimeters	25.13cm	$\text{Rev} = 100\text{cm}/25.12\text{cm}$ $\text{Rev} = 3.98\text{cm}$
100 inches	9.89in	$\text{Rev} = 100\text{in}/9.89\text{in}$ $\text{Rev} = 10.11\text{in}$
200 centimeters	25.13cm	$\text{Rev} = 200\text{cm}/25.13\text{cm}$ $\text{Rev} = 7.96\text{cm}$
200 inches	9.89in	$\text{Rev} = 200\text{in}/9.89\text{in}$ $\text{Rev} = 20.22\text{in}$
10 meters (1000 centimeters)	0.2513m 25.13cm	$\text{Rev} = 10\text{m}/0.2513\text{m}$ $\text{Rev} = 39.8\text{m}$
100 feet (1,200 inches)	0.82ft 9.89in	$\text{Rev} = 100\text{ft}/0.82\text{ft}$ $\text{Rev} = 122\text{ft}$

Radius (cm) r	Diameter (cm) d	Circumference $C = \pi * d$
8cm	16cm	50.26 cm

Distance d	Circumference (cm) C	Number of Revolutions $Rev = d/C$
100 centimeters	50.26 cm	$Rev = 100cm/50.26cm$ Rev=1.98cm
100 inches	19.78in	$Rev = 100in/19.78in$ Rev=5.055 in
200 centimeters	50.26cm	$Rev = 200cm/50.26cm$ Rev=3.97 cm
200 inches	19.78in	$Rev = 200in/19.78in$ Rev=10.11 in
10 meter (1000 centimeters)	0.5026 m	$Rev = 10m/0.5026m$ Rev=19.89m
100 feet (1,200 inches)	1.64 ft	$Rev = 100ft/1.64ft$ Rev=60.97 ft

Finally, the rolling friction was calculated. This was done to determine the amount of rolling friction acting against your mousetrap car and the coefficient of friction.

A moving mousetrap car is affected by two types of friction: air friction and bearing friction. To find the theoretical rolling friction requires placing your mousetrap car on a smooth and flat board or ramp. To calculate the theoretical rolling friction, a ramp is elevated from one end slowly until the mousetrap car begins rolling at a constant velocity. Using this data, the coefficient of friction can be found, as well as the force of friction.

Rolling Friction Data Table #1

Trial #	Board Length (m)	Raised Height (m)	Angle	Coefficient of Rolling Friction $\mu = f/mg$	Friction(N) $f = PE/d$	Starting Energy (J) mgh $M = .0924$	Predicted Travel Distance
1	L= 67.5cm 0.675m	h1= 2.7 cm 0.027 m	$\theta_1 = 0.04$	$\mu_1 = 1$	f1= 0.90552	PE=0 0.090552	d1=0.1m
2	L= 67.5cm 0.675m	h2= 3 2.8cm 0.028 m	$\theta_2 = 0.041$	$\mu_2 = 1$	f2= 0.90552	PE= 0.135828	d2= 0.15m
3	L= 67.5cm 0.675m	H3= 2.9 cm 0.029 m	$\theta_3 = 0.043$	$\mu_3 = 1$	f3= 0.90552	PE= 0.090552	d3= 0.1m

4	L= 67.5cm 0.675m	h4= 2.9 cm 0.029cm	θ4= 0.043	μ4= 1	f4= 0.90552	PE= 0.181104	d4=0.2m
AVE	L= 67.5cm 0.675m	- 0.02825m	θ= 0.04175	μ= 1	f= 0.90552	PE= 0.124509	d= 0.1375

To calculate the force of friction by assuming that the force down the ramp is equal to the force of friction acting against the motion of the vehicle. Using the equation weight equals mass times gravity (9.8 m/s²) as well as the potential energy calculated earlier. the predicted travel distance can be found by using the following formula: (Predicted Travel Distance = Total Potential Energy (joules or N-m)/(Rolling Friction))

Modified Predicted travel distance (PE/d)/(mgh)
Modified Predicted travel distance= 0.905/0.124 ≈ **7.27 m**

Analysis/Conclusion:

The data comparison between the basic mouse car trap and the modified mouse car trap is very promising in terms of reaching our goal of optimizing the mousetrap car to maximize distance and energy efficiency. To prove this, comparisons between the rolling friction calculated, mechanical advantage, and work done show that the mousetrap car improved due to the modifications made.

Rolling Friction:

Modifications we made to decrease the rolling friction and increase the predicted travel distance include adding bearings and sanding the car. To increase predicted travel distance, modifications such as cutting excess mass from the wheels and prioritizing weight reduction by laser cutting rectangles out of the body.

Calculated Values	Basic Mouse-Trap Car	Modified Mouse-Trap Car
Total Potential Energy	0.603 joules	0.905 joules
Rolling Friction	1.206 N	0.124 N
Predicted Travel Distance	2.00 m	7.27 m

Mechanical Advantage:

Modifications we made to increase the mechanical advantage include increasing wheel size of back wheels and increasing length of lever arm.

Mechanical Advantage	Basic Mouse-Trap Car	Modified Mouse-Trap Car
Lever Arm	7.5	9.64
Wheels and Axles	16	16 and 32

Work Done:

One modification we made was increasing the length of the lever arm. Based on the data collected originally, we understood that the longer the lever arm, the more work it was able to complete when attached to the spring.

Basic Mouse-Trap Car	Modified Mouse-Trap Car
1.005 Joules	1.29 Joules

In addition, the bottom strip, made from balsa wood, was super glued to the center of the car on the bottom to ensure and increase stability after cutting rectangles out of the base. The only inconsistency in the data which exists is the data found from wheel revolutions.

Wheel Revolutions:

One modification we made was increasing the size of the back wheels in the hopes of increasing mechanical advantage. This did happen, but as a result the revolutions for the back wheels decreased, which was not beneficial. However, it was determined that this loss of wheel revolution was worth the increased mechanical advantage of the wheel.

Wheel Revolutions (100cm distance)	Basic Mouse-Trap Car	Modified Mouse-Trap Car
Front Wheels	Rev: 3.98	Rev: 3.98
Back Wheels	Rev: 3.98	Rev: 1.98cm

Throughout this experience, lots was learned about the engineering design process. To talk about the engineering design process is one thing, but to work through it is another. During this process, the scientists ran into multiple problems. Oftentimes the car had too much friction or weighed too much and wouldn't even move. The team had multiple setbacks. Unrelated the engineering issues, an accident in the lab sometimes caused crucial pieces of the car, such as the mousetrap and the wheels, to break. If anything, the engineering design process has taught the engineers to be more creative and never give up.

In the introduction, it was mentioned that a deadly virus has spread throughout a hospital. All the patients, doctors, and occupants had been quarantined. Due to the virus being highly contagious, the easiest, safest, and most cost-efficient way to get the cure into the hospital it through mouse-trap cars. The car must travel long distances to carry to cure from outside the quarantine zone into the hospital. Following modifications made to the car into order to make it go faster, the mouse-trap car was able to transport the cure into the hospital, and the world was saved from a deadly virus outbreak.

Communicate

Through this engineering report, an in-depth discussion of the mouse car design project, as well as a presentation given on Monday, December 9th, as well as a demonstration of the car's ability to cover distant on Thursday, December 12th, the results of the engineering design process will be shared. The design project meets the requirements we set for it, meaning a 10% increase in distance is possible by changing the back wheels, material of the cars body, axel system and length of the lever arm of the mouse-trap car. To improve the car's performance in the future, the wheels should be aligned and fastened to provide stability, and everything should be sanded to reduce friction.

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